The Fine-Tuning Objection to Superdeterminism Conflates Specificity with Complexity

How Algorithmic Probability Dissolves the Primary Argument Against Violating Statistical Independence

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Abstract

The primary objection to superdeterministic interpretations of quantum mechanics is that reproducing the predictions of standard quantum theory requires implausibly specific, "fine-tuned", initial conditions correlating hidden variables with measurement settings. This objection conflates *specificity* with *complexity*. Under algorithmic probability, where structures are weighted by 2^(−K(S)) for Kolmogorov complexity K, initial conditions generated by simple rules have low descriptive complexity and are measure-favored, not measure-suppressed. We formalize the comparison by distinguishing three complexity quantities: the complexity of the laws K(L), of the initial conditions K(I), and of the specification of particular measurement outcomes K(O). Collapse interpretations incur outcome-specification costs K(O) that grow without bound; deterministic interpretations, both Many-Worlds and superdeterministic, avoid this cost entirely. The fine-tuning objection thus inadvertently argues against collapse, not against superdeterminism.

We engage with the strongest versions of the objection across the recent literature: Shimony, Horne & Clauser (1976) on the scientific method; Wiseman & Cavalcanti (2015) on implausibility; Maudlin (2014) on conspiracy; Baas & Le Bihan (2023) on Humean constraints and inductive generalization; Chen (2021) on measure-zero typicality; and Waegell & McQueen (2025) on the taxonomy of superdeterministic theories. We show that algorithmic probability provides a principled response to each. We propose a formal criterion distinguishing conspiratorial from non-conspiratorial correlations: a correlation is conspiratorial if and only if its Kolmogorov complexity exceeds that of the dynamics producing it. Dynamical attractor results (Ciepielewski, Okon & Sudarsky, 2023) and fractal invariant set analyses (Palmer, 2024) provide independent evidence that superdeterministic correlations satisfy this criterion. MWI retains a parsimony advantage with current models, but the competition is mathematical, which generating rule has lower total K, not settled by appeals to intuitive plausibility.

Keywords: superdeterminism, statistical independence, Bell's theorem, Kolmogorov complexity, algorithmic probability, fine-tuning, measurement prob-

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1. Introduction

Bell's theorem (1964) demonstrates that no local hidden-variable theory satisfying statistical independence (SI), the assumption that measurement settings are uncorrelated with the hidden variables governing outcomes, can reproduce all predictions of quantum mechanics. The theorem permits three responses: accept nonlocality, reject hidden variables, or reject SI. The third option, superdeterminism, holds that in a fully deterministic universe, measurement settings and hidden variables share causal ancestry in the initial conditions, permitting local explanations of Bell correlations.

The mainstream physics community largely dismisses superdeterminism. The objections cluster into three families:

(O1) Fine-tuning. Reproducing quantum predictions requires implausibly specific initial conditions encoding correlations between hidden variables and every possible future measurement configuration (Zeilinger, 1999; Wiseman & Cavalcanti, 2015; Baas & Le Bihan, 2023).

(O2) Scientific method. If measurement settings are correlated with hidden variables, controlled experiments, and therefore the scientific method itself, become impossible (Shimony, Horne & Clauser, 1976; Maudlin, 2014; Chen, 2021).

(O3) Model deficit. No complete superdeterministic model reproducing all predictions of quantum field theory exists (Araújo, 2019; Waegell & McQueen, 2025).

This paper addresses (O1) directly, shows that addressing (O1) partially dissolves (O2), and acknowledges (O3) as an open problem that does not bear on the logical status of (O1). The argument proceeds from algorithmic probability, specifically, the application of Solomonoff's (1964) theory of inductive inference to questions of physical ontology, an extension developed by Schmidhuber (1997) and adapted here as algorithmic probability (Solomonoff, 1964; Schmidhuber, 2000).

The central claim is that the fine-tuning objection conflates two distinct questions:

- **(Q1)** Are the required initial-condition correlations *specific*? (Yes, necessarily.)
- **(Q2)** Are the required initial-condition correlations *complex*? (Unknown, depends entirely on whether they follow from a simple generating rule.)

Answering (Q1) affirmatively does not entail an affirmative answer to (Q2). Conflating these questions is a category error analogous to calling the digits of "fine-tuned" because they are specific.

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2. Formal Framework

2.1 Definitions

Let S denote a complete physical theory comprising dynamical laws L, initial conditions I, and (where applicable) a specification of which particular measurement outcomes occurred, O. The Kolmogorov complexity K(S) is the length of the shortest program that outputs S on a universal Turing machine U. Although K is U-dependent up to an additive constant, the *relative* complexities K(S) − K(S) are well-defined and U-independent for sufficiently complex structures.

We distinguish three complexity quantities:

K(L): The complexity of the dynamical laws alone. For standard quantum mechanics with collapse, this includes the Schrödinger equation plus the measurement postulate (Born rule (that probabilities follow the square of the wave amplitude) and projection). For MWI, only the Schrödinger equation. For a superdeterministic theory, the dynamical law plus whatever rule generates the SI-violating correlations.

K(I): The complexity of the initial conditions. For a universe whose initial state follows from a compact generating function, K(I) is low, bounded by the length of that function. For a universe whose initial state is a brute fact requiring element-by-element specification, K(I) is high, potentially proportional to the number of degrees of freedom.

K(O): The complexity of specifying which particular outcomes occurred at each measurement event. Under collapse interpretations, each measurement selects one eigenvalue (specific measurement outcome) from the Born-rule probability distribution. This selection is additional information not determined by the wave function. Over the history of a universe with N measurement events, K(O) grows with N.

2.2 Algorithmic probability

Under algorithmic probability, adapted from Solomonoff's (1964) universal prior over computable hypotheses and extended from epistemology to physical ontology by Schmidhuber (1997):

> The probability of a physical structure S obtaining is P(S) 2^(−K(S)), where K(S) = K(L) + K(I) + K(O).

This postulate is not new physics. It is the application of a well-established principle in algorithmic information theory, that simpler descriptions are exponentially more probable under any universal prior, to the question of which physical theories describe actuality. The justification is the same as Solomonoff's: among all computable structures consistent with observations, simpler ones dominate the measure.

2.3 The Born Rule Objection and Its Resolution

A natural objection arises: quantum mechanics specifies the probability distribution over outcomes with finite K (the Born rule $| |^2$ is a compact postulate). Only specifying *which particular* outcome occurred requires additional K. So perhaps K(O) is irrelevant, the *theory* has finite K regardless.

This objection conflates the complexity of the predictive theory with the complexity of the actual physical history. The Born rule specifies a distribution; it does not specify which branch of the distribution is actualized. Under collapse interpretations, the universe must select one outcome at each measurement. That selection is, by hypothesis, irreducibly random, not determined by the wave function, the Born rule, or anything else in the theory. The sequence of selections across all measurement events in the universe's history constitutes an algorithmically incompressible string: K(O)   N for N measurements with random outcomes.

This is not a deficiency of collapse interpretations that might be repaired. It is the defining feature: collapse theories hold that nature makes choices not determined by the prior state. Each choice adds information, specification complexity, to the complete description of the physical world.

Under MWI, all outcomes occur. No selection is made. K(O) = 0. The theory's total complexity is K(L_MWI) + K(I). There is a cost, the ontological profligacy of the branching multiverse, but this cost is not a *K*-cost. The Schrödinger equation generates all branches from one specification.

Under superdeterminism, outcomes are determined by the initial conditions. K(O) = 0, because outcomes are computable from L + I. There is a cost, the initial conditions may be complex, but this cost is K(I_SD), which is bounded and depends on the generating rule.

2.4 The K-Cost Comparison

The total specification complexity for each interpretive family is:

- **Collapse:** K(L_collapse) + K(I) + K(O), where K(O) grows without bound
- **MWI:** K(L_Schrödinger) + K(I), with K(O) = 0
- **Superdeterminism:** K(L_SD) + K(I_SD), with K(O) = 0

Both MWI and superdeterminism avoid the unbounded K(O) cost. The fine-tuning objection claims K(I_SD) is large. We now argue this claim is unwarranted.

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3. The Category Error

3.1 Specificity versus Complexity

The fine-tuning objection assumes that SI-violating initial conditions must be specified element-by-element. For every future measurement event, the initial state must encode the correlation between the hidden variable and the measurement setting (a, b). If there are N such events over the history of the universe, this element-by-element specification yields K(I_SD) N, enormous, and potentially larger than K(O) under collapse.

Under this treatment, the objection holds: brute-fact correlations are maximally complex and measure-suppressed under algorithmic probability.

But the treatment assumes that the correlations *are* brute facts, an arbitrary lookup table mapping measurement events to hidden-variable states. This is the unexamined assumption at the heart of the objection.

The alternative: the SI-violating correlations follow from a *simple generating rule*, a compact function that produces the initial state, from which all future correlations follow deterministically. Under this alternative, K(I_SD) is the length of the generating function, which may be small.

The digits of illustrate the distinction. The first billion digits of are extraordinarily specific: altering any single digit produces a different (and non-) sequence. A critic who examined these digits without knowing their source might call them "fine-tuned", they appear random, they are highly specific, and they admit no obvious pattern. Yet K( ) is small, because a program of a few dozen bytes generates all of them. *Specificity is not complexity.*

The fine-tuning objection observes that SI-violating correlations are specific, they must be exactly right to reproduce quantum predictions, and concludes they are complex. This is the category error. Whether the correlations are complex depends on whether they follow from a simple rule, which is an open empirical and theoretical question, not something settled by the specificity alone.

3.2 Dynamical Attractor Support

Ciepielewski, Okon, and Sudarsky (2023) provide independent evidence against the brute-fact assumption. Working within a superdeterministic variant of Bohmian mechanics (Leibnizian Quantum Mechanics, or LQM), they demonstrate that the SI-violating correlations arise as a *dynamical attractor*. Starting from generic, not fine-tuned, initial conditions, the dynamics alone drive the system toward the homogeneous state that correlates hidden variables with measurement settings.

Under algorithmic probability, this result has a precise interpretation: if the dynamics have low K(L), and the correlated state is their attractor, then the correlations inherit the simplicity of the dynamics. K(I_SD) is not an independent cost to be added to K(L_SD), it is *absorbed into* K(L_SD). The attractor does the work that the fine-tuning objection claims requires brute specification.

Note that Ciepielewski et al. do not claim LQM is a serious competitor to standard quantum mechanics, they are explicit that it is a toy model. The point is structural: if SI-violating correlations *can* arise as attractors of simple dynamics, then the assumption that they *must* be brute facts is falsified by counterexample.

3.3 Fractal Invariant Set Support

Palmer (2024) arrives at a complementary conclusion through a different mathematical route. Working with p-adic metrics on fractal invariant sets in state space, Palmer argues that perturbations taking the system off the invariant set are *large* in the natural p-adic metric of state space, even when they appear small in the Euclidean metric. What looks fine-tuned from the Euclidean perspective is generic from the p-adic perspective.

This dovetails with the algorithmic probability argument: the "naturalness" of initial conditions is measure-dependent. The Euclidean measure, which the fine-tuning objection implicitly assumes, treats nearby states as similar regardless of dynamical consequences. The p-adic measure treats states as similar only if they share dynamical trajectories. Under the latter, the invariant set is not special; it is the natural home of the dynamics.

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4. Engaging the Strongest Objections

4.1 Shimony, Horne & Clauser (1976): The Scientific Method

The foundational objection: "In any scientific experiment in which two or more variables are supposed to be randomly selected, one can always conjecture that some factor in the overlap of the backwards light cones has controlled the presumably random choices. But, we maintain, skepticism of this sort will essentially dismiss all results of scientific experimentation."

This objection targets the scientific method, not fine-tuning per se. Its force derives from the assumption that SI is necessary for experimental inference. Under algorithmic probability, the scientific method works not because experimenters possess libertarian free will, they do not under *any* deterministic interpretation, including MWI, but because simple-law structures produce regularities that are reliably discoverable through observation. The correlations between what we choose to measure and what we find are structural features of simple dynamics, not obstacles to knowledge.

The practical success of science is explained by the same parsimony that explains why our laws are simple: algorithmic probability weights simple-law structures highest, and simple laws generically produce statistical regularities that hold across sub-ensembles, i.e., they produce effective SI at the macroscopic level even if SI is violated at the fundamental level.

4.2 Wiseman & Cavalcanti (2015): Implausibility

Wiseman and Cavalcanti argue that any superdeterministic theory "would be about as plausible, and appealing, as belief in ubiquitous alien mind-control." The rhetorical force rests on the intuition that SI-violating correlations are implausible, cosmically unlikely.

But plausibility is *measure-dependent*. Under the uniform measure over initial conditions (which the objection implicitly assumes), SI-violating correlations are indeed atypical: a randomly chosen initial state will generically satisfy SI. Under algorithmic probability, the relevant measure is not uniform but complexity-weighted: the most probable initial states are those generated by the shortest programs. Whether SI-violating correlations are typical or atypical under this measure depends on whether the generating rule is simple.

The objection is correct under its assumed measure and inapplicable under algorithmic probability. The question of which measure is appropriate for fundamental physics is itself a substantive question, one that algorithmic probability, with its foundations in Solomonoff induction and Kolmogorov complexity, is well positioned to address.

4.3 Maudlin (2014): Conspiracy

Maudlin's version of the objection emphasizes the "conspiratorial" character of SI-violating correlations: nature must arrange, in advance, for every measurement outcome to be correlated with every measurement choice.

Palmer (2024) has already shown that while conspiracy implies SI-violation, the converse is false: superdeterminism does not logically entail conspiracy. The present paper adds a *formal* criterion:

> **Definition.** A correlation between variables X and Y is *conspiratorial* relative to dynamics D if and only if K(correlation | D) is large, i.e., specifying the correlation requires substantial information beyond specifying the dynamics.

Under this criterion:

- If SI-violating correlations are brute facts (lookup tables), K(correlation | D) N. This is conspiratorial.
- If SI-violating correlations are attractors of the dynamics, K(correlation | D) 0. This is not conspiratorial, the dynamics produce the correlations inevitably.
- If SI-violating correlations are structural features of a fractal invariant set defined by the dynamics, K(correlation | D) 0 for the same reason.

The fine-tuning objection assumes the first case. Ciepielewski et al. (2023) and Palmer (2024) provide existence proofs for the second and third. The question of which case obtains in the actual universe is empirical and theoretical, not settled by the objection itself.

4.4 Baas & Le Bihan (2023) and Chen (2021)

Baas and Le Bihan (2023) argue that SI-violation implies experimental samples are not representative of populations, undermining inductive generalization. Their deeper move is that superdeterminism makes the coordination between laws and initial conditions suspicious, it requires a Humean account where both are contingent and mysteriously aligned.

Under mathematical structural realism, this suspicion dissolves: laws and initial conditions are both outputs of the same generating rule. The coordination is not mysterious, it is what simple generating functions do. The rule produces the dynamics AND the initial state, the way a short program produces both the algorithm and its input.

Chen (2021) offers one of the most careful philosophical formulations of the fine-tuning objection, arguing that superdeterministic initial conditions require measure-zero fine-tuning under the natural (Lebesgue) measure. The algorithmic probability response is direct: the Lebesgue measure is not the correct measure over physical structures. Algorithmic probability weights structures by generative simplicity, not by volume in configuration space. A measure-zero set under Lebesgue can have high algorithmic probability if it is the output of a short program. This is the core of the category error in one sentence: the fine-tuning objection uses the wrong measure.

On inductive generalization: SI is violated in principle (hidden variables are correlated with settings) but may be valid for practical purposes, a distinction Andreoletti and Vervoort (2022) have independently defended. Hossenfelder and Palmer (2020) argue the deviations from quantum statistics are analogous to climate versus weather: real but too small for current instruments to resolve. Whether this is correct is an empirical question that depends on model-specific predictions not yet available. We do not claim to have resolved the practical-SI question. The point is that the objection does not succeed as a *principled* argument against superdeterminism; it identifies a *difficulty* that any future superdeterministic model must address.

4.5 Waegell & McQueen (2025): The Taxonomy

Waegell and McQueen provide the most comprehensive recent analysis of the superdeterminism landscape, classifying SI-violating theories into three categories:

1. **Deterministic theories with fine-tuned initial conditions.** The hidden variables and settings are correlated because the initial state was set up that way.

2. **Fluke theories (statistical flukes).** The correlations arise by chance from typical initial conditions.
3. **Nomic exclusion theories.** The laws of physics exclude certain measurement-setting/hidden-variable combinations from the space of physically possible states.

They conclude that superdeterminism is "conspiratorial but not unscientific, pre-scientific."

The algorithmic probability argument applies differently to each category. For Category 1, algorithmic probability directly dissolves the fine-tuning charge: if the initial conditions follow from a simple generating rule, they are not fine-tuned in the K-complexity sense. For Category 2, algorithmic probability is neutral, statistical flukes have high K by definition. For Category 3, algorithmic probability provides positive support: if the laws are simple (low K(L)), and simple laws define an invariant set that excludes certain states, the exclusion is structural and inherits the simplicity of the laws.

The Waegell-McQueen conclusion that superdeterminism is "conspiratorial" rests on a definition of conspiracy that is independent of the generating mechanism. Under our formal criterion (§4.3), whether superdeterminism is conspiratorial depends on K(correlation | dynamics). The taxonomy is useful; the blanket "conspiratorial" verdict does not survive algorithmic analysis. Categories 1 (with simple generating rules) and 3 need not be conspiratorial under our criterion.

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5. MWI versus Superdeterminism: A Complexity Competition

Both MWI and superdeterminism are fully deterministic. Neither incurs the unbounded K(O) cost of collapse. The competition between them reduces to:

- K(MWI) = K(Schrödinger equation) + K(I)
- K(SD) = K(L_SD) + K(I_SD)

If the left-hand side is smaller, MWI dominates on parsimony. If the right-hand side is smaller, superdeterminism dominates.

With current models, MWI retains a parsimony advantage. The Schrödinger equation alone, with no additional postulates, has minimal K among known quantum frameworks. No superdeterministic model has yet demonstrated competitive total K. This is a contingent fact about the current state of model-building, not a principled impossibility.

However, the naive comparison K(MWI)   K(Schrödinger equation) may undercount MWI's actual complexity costs. MWI faces two unresolved problems that may represent hidden K:

The Born rule derivation problem. MWI must derive the Born rule (probabilities follow $| |^2$) rather than postulate it. Deutsch (1999) and Wallace (2012) argue it follows from decision-theoretic rationality constraints. Whether this derivation succeeds is contested (see Kent, 2010, for objections). If an additional postulate is required to recover the Born rule, K(MWI) increases by the complexity of that postulate.

The preferred basis problem. MWI must explain why decoherence selects the basis we observe (position, not momentum). If decoherence is sufficient, as Wallace (2012) argues, no additional K is needed. If decoherence requires supplementary assumptions about the structure of the Hamiltonian or the initial state, further specification complexity is incurred.

These are not fatal to MWI's parsimony advantage, they are open questions. But they mean the K-comparison is less lopsided than it first appears. The competition is mathematical, open, and may be closer than the standard narrative suggests.

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6. Connections to Existing Complexity Arguments

Hossenfelder and Palmer (2020) already gesture toward complexity-based reasoning in the superdeterminism literature. They define scientific explanation as the ability to "calculate measurement outcomes in a way that is computationally simpler than just collecting the data," and note that Chaitin's incompleteness theorem is relevant: it is provably unprovable whether a given theory's initial conditions are simpler than the data it explains. This is the complexity landscape in which our argument operates. More broadly, Hossenfelder (2018) asks why physics should prefer simple or beautiful theories, and finds no satisfactory answer. Algorithmic probability provides one: lower-K structures have exponentially higher measure under algorithmic probability. The criterion is not beauty but brevity, and brevity has measurable consequences for measure. That physicists are drawn to calling low-K solutions elegant may itself be adaptive. The criterion remains mathematical.

What the present paper adds to this landscape is fourfold: (a) a formal decomposition of total complexity into K(L) + K(I) + K(O), which makes the comparison between interpretive frameworks precise; (b) the observation that collapse theories incur unbounded K(O) costs that deterministic theories avoid, a point that, to our knowledge, has not been made explicitly in the superdeterminism literature; (c) a formal criterion for conspiracy based on conditional Kolmogorov complexity K(correlation | dynamics), which replaces intuitive plausibility judgments with a mathematical criterion; and (d) the explicit connection to Solomonoff's universal prior, which provides algorithmic probability with established foundations in algorithmic information theory rather than treating it as a novel, unsupported postulate.

The same K-decomposition yields a further result: the thermodynamic arrow of time. A universe described forward from initial conditions I has total complexity K(L) + K(I), both low. Described backward from final state I_f, the total is K(L) + K(I_f), where K(I_f) encodes the accumulated computational output of the evolution, enormous and potentially incompressible. Under algorithmic probability, forward-described structures dominate exponentially. The Past Hypothesis (Albert, 2000) follows as theorem rather than brute postulate (Bernstein, 2026e).

Adlam (2021) provides independent support from a different direction. Her analysis of contextuality and fine-tuning argues that what appears fine-tuned under one measure may be natural under another, a conclusion that converges with the present K-based criterion. The fine-tuning charge against superdeterminism assumes a background measure (typically Lebesgue on state space) and finds the required correlations unlikely. But as Adlam shows, the choice of measure is itself a substantive physical commitment. The algorithmic probability criterion formalizes this: fine-tuning is high conditional K relative to the dynamics, not improbability under an externally imposed measure. Adlam, Hossenfelder, Hance, and Palmer (2023) further clarify the taxonomy of SI-violating models, distinguishing superdeterministic, retrocausal, and atemporal approaches. The present argument applies across their taxonomy: any SI-violating model whose total K is lower than collapse alternatives is favored under algorithmic probability, regardless of which category it occupies.

7. Conclusion

The extreme fine-tuning objection to superdeterminism rests on the implicit, unexamined assumption that SI-violating initial conditions are brute facts with high Kolmogorov complexity. Under algorithmic probability, initial conditions generated by simple rules have low K and are measure-favored. The objection conflates specificity with complexity, a category error.

Four independent lines of evidence support this conclusion:

(1) The formal K-decomposition (§2) shows that collapse, not superdeterminism, incurs the only provably unbounded complexity cost (K(O)).

(2) Dynamical attractor results (Ciepielewski et al., 2023) demonstrate that SI-violating correlations can arise inevitably from generic initial conditions under simple dynamics.

(3) Fractal invariant set analysis (Palmer, 2024) shows the “fine-tuned” state is generic in the natural topology of state space.

(4) The formal conspiracy criterion (§4.3) distinguishes high-K brute-fact correlations from low-K dynamically-generated correlations, removing the rhetorical force of the “conspiracy” charge.

MWI currently leads on parsimony. The competition is mathematical, open, and requires knowing both generating rules to adjudicate. But the fine-tuning objection, the argument most commonly deployed to dismiss superdeterminism without engaging its substance, is itself based on a category error. The field should evaluate superdeterministic models on their Kolmogorov complexity, not on intuitive plausibility judgments that are artifacts of an assumed measure.

A clarification on scope: the present analysis is compatible with both MWI and superdeterminism. The claim is that the fine-tuning objection fails on its own terms, not that superdeterminism is the correct interpretation. Readers who find MWI's parsimony decisive may agree with every argument in this paper while maintaining their preference. The contribution is to remove a bad argument from a legitimate debate, not to settle the debate itself.

A further philosophical observation: true randomness, like a perfect circle, is definable but not constructible. Any process that produces an output is itself structured, rendering the output determined by that structure. If genuine randomness is unconstructible, superdeterminism is not merely one interpretation among several but the only constructible one.

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